THE DERIVED LIMIT OF AN INVERSE SYSTEM OF PERFECT COMPLEXES ALONG SURJECTIONS WITH LOCALLY NILPOTENT KERNELS IS PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, together with the project's own definitions of the reference/slogan/history display environments, which the project itself typesets as comments; nothing was added, removed or redefined by this card and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted statement span includes the author's own \begin{reference} block citing the source of the argument; it is delivered verbatim and produces no unresolved citation. The five references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- The pseudo-coherent version of the statement, giving pseudo-coherence and the base change isomorphisms. invoked at proof lines 28422-28424. Provides the starting point from which perfection is upgraded.
- Elements mapping to units are units, and the resulting containment in the Jacobson radical. used at proof lines 28427-28430. Is what lets a surjection to a residue field factor through A_1.
- Perfect complexes have finite tor dimension, and the pointwise criterion for perfection. invoked at proof lines 28438-28443. Convert the boundedness of the residue field fibres into perfection of K.

Source reference. Original lemma 0AME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13402–13402. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0AMG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13404–13404. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 28336: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28336–28336. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 20032: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20032–20032. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18889: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18889–18889. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A = \lim A_n$ be a limit of an inverse system (A_n) of rings. Suppose given $K_n \in D(A_n)$ and maps $K_{n+1} \to K_n$ in $D(A_{n+1})$. Assume*

1. *the transition maps $A_{n+1} \to A_n$ are surjective with locally nilpotent kernels,*
2. *either all K_n are perfect or there exists an integer n_0 such that K_{n_0} is perfect and the kernels $A_{n+1} \to A_n$ are nilpotent for $n \geq n_0$, and*
3. *the maps induce isomorphisms $K_{n+1} \otimes^{\mathbf{L}}_{A_{n+1}} A_n \to K_n$.*

Then $K = R\lim K_n$ is a perfect object of $D(A)$ and $K \otimes^{\mathbf{L}}_A A_n \to K_n$ is an isomorphism for all n.

Proof. We already know that K is pseudo-coherent and that $K \otimes^{\mathbf{L}}_A A_n \to K_n$ is an isomorphism for all n by Lemma 28336. Consider a surjective map $A \to \kappa$ whose kernel is a maximal ideal $\mathfrak{m}$. Any element of A which maps to a unit in A_1 is a unit in A by Algebra, Lemma 0AMG and hence $\mathrm{Ker}(A \to A_1)$ is contained in the Jacobson radical of A by Algebra, Lemma 0AME. Hence $A \to \kappa$ factors as $A \to A_1 \to \kappa$. Hence

$$K \otimes^{\mathbf{L}}_A \kappa = K \otimes^{\mathbf{L}}_A A_1 \otimes^{\mathbf{L}}_{A_1} \kappa = K_1 \otimes^{\mathbf{L}}_{A_1} \kappa$$

Note that K_1 is perfect by assumption (2). Hence K_1 has finite tor dimension by Lemma 18889. Thus there exist $a, b \in \mathbf{Z}$ such that $H^i(K \otimes^{\mathbf{L}}_A \kappa) = 0$ for all $i \notin [a, b]$. By Lemma 20032 we conclude that K is perfect. □